Permutation of repeated things:

The number of permutation of n different things taken r at a time, when repetition is allowed is n^r.

Example 2.20

Using 9 digits from 1, 2, 3, … , 9 taking 3 digits at a time, how many 3 digits numbers can be formed when repetition is allowed?

Solution

Here, $n = 9$ and $r = 3$

$\therefore$ Number of 3 digit numbers = n^r

$= 9^3 = 729$ numbers

Permutations when all the objects are not distinct:

The number of permutations of n things taken all at a time, of which p things are of one kind and q things are of another kind, such that $p + q = n$ is $\frac{n!}{p!\,q!}$.

In general, the number of permutation of n objects taken all at a time, p_1 are of one kind, p_2 are of another kind, p_3 are of third kind, … p_k are of kth kind such that $p_1 + p_2 + \ldots + p_k = n$ is $\frac{n!}{p_1!p_2!\ldots p_k!}$.

Example 2.21

How many distinct words can be formed using all the letters of the following words.

(i) MISSISSIPPI (ii) MATHEMATICS.

Solution

(i) There are 11 letters in the word MISSISSIPPI

In this word M occurs once

I occurs 4 times

S occurs 4 times

P occurs twice.

Therefore required number of permutation = $\frac{11!}{4!\,4!\,2!}$

(ii) In the word MATHEMATICS

There are 11 letters

Here M occurs twice

T occurs twice

A occurs twice

and the rest are all different. Therefore number of permutations = $\frac{11!}{2!\,2!\,2!}$

2.2.5 Circular permutation

In the last section, we have studied permutation of n different things taken all together is $n!$. Each permutation is a different arrangement of n things in a row or on a straight line. These are called Linear permutation. Now we consider the permutation of n things along a circle, called circular permutation.

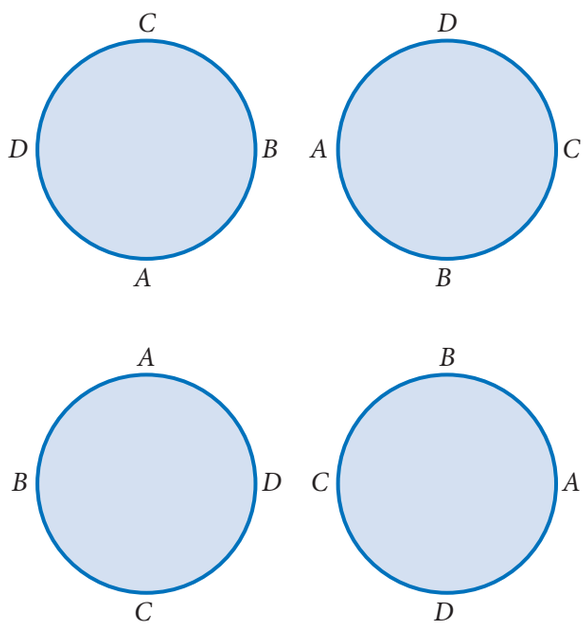


Fig. 2.4

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